

AI Fintech Super Platform - Detailed Graphical Architecture Flow

This document provides a detailed graphical architecture representation for the AI-powered fintech super platform. It includes end-to-end architecture flow, AI recommendation pipeline, realtime communication architecture, deployment flow, event-driven architecture, and security design.

1. High-Level Platform Architecture

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graph TD
    A[CLIENT APPLICATIONS] --> B[API GATEWAY]
    B --> C[MICROSERVICES LAYER]
    C --> D[EVENT-DRIVEN MESSAGING LAYER]
    D --> E[AI + RAG LAYER]
    E --> F[DATA LAYER]
    F --> G[REALTIME COMMUNICATION LAYER]
    G --> H[USER DASHBOARD]
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2. Client Applications Layer

- React Web Application
- Progressive Web App (PWA)
- Mobile App (Future)
- AI Financial Chat Assistant
- Admin Dashboard
- SignalR Realtime Connections

3. API Gateway Layer

Responsibilities:

- Authentication validation
- API routing
- Request aggregation
- Rate limiting
- Logging and monitoring
- Response caching
- Circuit breaker implementation

4. Core Microservices Architecture

Identity Service:

- JWT authentication
- MFA validation
- Session management

Portfolio Aggregation Service:

- Holdings synchronization
- Transaction analysis
- PnL calculations

Broker Connector Service:

- Zerodha API
- Groww API
- Angel One API
- Upstox API
- ICICI Direct API

AI Recommendation Service:

- AI prompts orchestration
- RAG processing
- Portfolio analysis
- AI scoring

News Intelligence Service:

- Realtime news ingestion
- NLP sentiment analysis
- Market event detection

5. Event-Driven Messaging Architecture

Kafka Topics:

- portfolio-updated
- market-news-received
- ai-analysis-generated
- forex-rate-updated
- commodity-alert-generated

RabbitMQ Queues:

- AI background jobs
- Notification jobs
- PDF parsing jobs
- Broker synchronization jobs

6. AI + RAG Architecture

User Query

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Embedding Generation

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Qdrant Vector Search

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Context Retrieval

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Semantic Kernel Orchestration

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OpenAI LLM

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AI Recommendation Generation

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Dashboard Delivery

Knowledge Sources:

- NSE/BSE data
- SEBI circulars
- RBI updates
- Company annual reports
- Economic indicators
- Global market news

7. Realtime Communication Flow

SignalR Hubs:

- portfolioHub
- marketHub
- AIHub
- notificationHub

Realtime Features:

- Live portfolio updates
- AI alerts
- Forex alerts
- Commodity alerts
- Market event streaming
- Breaking news notifications

8. Data Layer Architecture

Primary Database:

- PostgreSQL

Caching:

- Redis cache
- AI response caching
- Session caching

Vector Database:

- Qdrant

Blob Storage:

- Azure Blob Storage

9. Security Architecture

Security Layers:

- TLS 1.3 transport encryption
- AES-256 encrypted storage
- JWT authentication
- RBAC authorization
- Device fingerprinting
- Refresh token rotation
- Secure broker token encryption
- Signal protocol-based E2EE

10. Deployment Architecture

Developer Push

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GitHub Actions

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Docker Build Pipeline

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Automated Testing

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Azure Container Registry

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Azure Container Apps
↓
Canary Deployment
↓
Production Environment

Monitoring Stack:

- OpenTelemetry
- Grafana
- Azure Application Insights

11. Dashboard Modules

Dashboard Features:

- Portfolio Analytics
- AI Financial Assistant
- MCX Commodity Insights
- Forex Analytics
- Goal Planning
- Retirement Planner
- AI News Intelligence
- Notifications Center

12. Recommended Technology Stack

Frontend:

- React 19
- TypeScript
- Tailwind CSS
- Redux Toolkit

Backend:

- ASP.NET Core 9
- Clean Architecture
- CQRS
- Mediatr

AI Stack:

- OpenAI APIs
- Semantic Kernel
- LangChain
- Qdrant

Infrastructure:

- Azure Container Apps
- PostgreSQL
- Redis
- Kafka
- RabbitMQ
- SignalR

13. End-to-End User Flow

Step	Description
1	User logs into platform
2	API Gateway validates request
3	Identity Service generates JWT token
4	Broker account connected

5	Portfolio synchronization starts
6	Kafka events published
7	AI recommendation pipeline triggered
8	Realtime SignalR updates sent
9	Dashboard refreshed dynamically

14. Conclusion

Final Recommendation: This AI fintech super platform architecture is designed for scalability, security, realtime analytics, AI-native financial intelligence, and production-grade cloud deployment. The architecture combines microservices, event-driven systems, AI orchestration, realtime communication, and secure fintech workflows.